

🔧 AWS CLF-C02 — Task 1.1 Final Exam Tips

1 AWS Cloud Value Proposition

- **Key Idea:** AWS = **faster, cheaper, scalable, secure, global**.
 - **Memory Hook:** "Speed + Cost + Scale + Security + Global reach"
 - **Example:** Netflix deploys streaming services globally without owning servers.
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2 Benefits of Cloud Computing

- **Agility:** Launch apps quickly → *EC2, Lambda, Elastic Beanstalk*
- **Elasticity:** Auto scale → *EC2 Auto Scaling, Lambda*
- **Scalability:** Handle growth → *S3, RDS, DynamoDB*
- **Reliability:** Multi-AZ, fault-tolerant → *RDS Multi-AZ, S3 durability 11 nines*
- **Security:** Encryption, IAM, compliance → *IAM, KMS, Shield*
- **Global Reach:** Low latency worldwide → *CloudFront, multiple Regions*

Memory Trick: "A E S R S G" → Agility, Elasticity, Scalability, Reliability, Security, Global Reach

3 AWS Global Infrastructure

- **Region:** Geographic area → choose based on latency/compliance
- **AZ (Availability Zone):** Independent data centers → high availability
- **Edge Location:** CDN → low-latency content delivery
- **Local Zone:** Close to cities → ultra-low latency for apps
- **Outposts:** AWS on-premises → hybrid cloud for sensitive data

Memory Hook: "Region → AZ → Local Zone → Edge → Outpost"

4 Economies of Scale & Cost Savings

- AWS passes **massive infrastructure cost savings** to customers.
- Pay-as-you-go pricing → no upfront CapEx.
- Use **Spot Instances, Auto Scaling, Right-sizing** for cost optimization.

Tip: When a question mentions "lower costs", think **Economies of Scale + Pay-as-you-go + Cost optimization**.

5 Shared Responsibility Model (High-Level)

- **AWS:** Security **of** the cloud → hardware, network, managed services
- **Customer:** Security **in** the cloud → data, apps, OS, IAM

Memory Hook: “AWS = cloud; You = in cloud”

6 How AWS Enables Innovation & Digital Transformation

- Rapid prototyping → Lambda, API Gateway
- Access to advanced tech → SageMaker, IoT, Analytics
- Scalability & elasticity → Auto Scaling, ELB
- Faster time-to-market → Regions + Edge Locations
- Cost-efficient experimentation → Pay-as-you-go
- Hybrid & modernization → Outposts, Local Zones

Tip: Keywords in exam: “**innovation, experiment, speed, scale, modernize**” → AWS answer

7 Skills Focus for Exam

- **Explain benefits vs traditional data centers:**
 - AWS = no upfront hardware, global deployment, elastic resources
 - **Identify global infrastructure advantages:**
 - High availability, low latency, fault tolerance
 - **Describe elasticity and scalability:**
 - Auto Scaling, serverless (Lambda), pay-for-what-you-use
 - **Explain pay-as-you-go pricing:**
 - Only pay for what you use → CapEx vs OpEx
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⚡ Final Memory Hack for Task 1.1

Mnemonic:

"V-B-G-E-S-I"

V = Value proposition

B = Benefits (Agility, Elasticity, Scalability, Reliability, Security, Global Reach)

G = Global Infrastructure (Regions, AZ, Local Zones, Edge, Outposts)

E = Economies of scale & cost savings

S = Shared Responsibility Model

I = Innovation & digital transformation

Think of this **every time you see Task 1.1** — it covers **all knowledge and skills** for the exam.

🔧 Task 1.2 — AWS Cloud Design Principles: Final Exam Tips

1 The 6 Design Principles (Well-Architected Framework)

Principle	Key Idea	Real-World Example	Exam Tip / Memory Hook
Design for Failure	Assume components fail → build auto-recovery	Multi-AZ RDS continues serving even if one AZ fails	“Expect failure → recover automatically”
Implement Elasticity	Scale resources up or down dynamically	EC2 Auto Scaling during Black Friday sales	Elasticity = dynamic scaling based on demand
Leverage Managed Services	Use AWS-managed services instead of building infrastructure	Amazon RDS instead of self-managed databases	Managed = less maintenance, focus on apps
Automate Everything	Automate deployments, monitoring, updates	AWS CloudFormation deploys infra as code	Automation = faster, fewer errors
Think Parallel	Run tasks concurrently for speed and efficiency	EMR / Lambda parallel data processing	Parallel = faster results, scalable processing
Stop Guessing Capacity Needs	Don't over-provision → pay only for what you use	Auto Scaling + S3 storage	Pay-as-you-go → OpEx vs CapEx

Mnemonic: F-E-M-A-P-C → **Failure, Elasticity, Managed, Automate, Parallel, Capacity** (“FEM A PC”)

2 High Availability & Fault Tolerance

- **High Availability:** Keep applications **up and running** during failures.
 - **Fault Tolerance:** System continues working **even if components fail**.
 - **Use Cases:**
 - **Multi-AZ deployment** → database or app spans multiple AZs in a Region
 - **Multi-Region deployment** → global apps survive a Region outage
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3 Skills for the Exam

- **Recognize best practices:**
 - Use **managed services** for reliability
 - **Automate deployments** → fewer manual errors
 - **Elastic & parallel workloads** → cost-efficient and fast

- **Identify when to use Multi-AZ vs Multi-Region:**
 - **Multi-AZ:** Within a single Region → HA and fault tolerance
 - **Multi-Region:** For disaster recovery or global apps → resilience and low latency
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4 Quick Exam Memory Tricks

1. **Mnemonic:** F-E-M-A-P-C → easy recall of 6 principles
 2. **HA vs Fault Tolerance:** HA = **up time**, FT = **continue working if failure**
 3. **Elasticity:** Auto-scale → think **Black Friday traffic**
 4. **Parallel:** Run tasks concurrently → **big data / Lambda**
 5. **Capacity:** Don't over-provision → pay only for what you use
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☑ Pro Tip:

- Exam questions often phrase scenarios like:
 - *“Which principle ensures apps remain online if one data center fails?”* → *Design for Failure*
 - *“Which approach avoids over-provisioning?”* → *Stop Guessing Capacity Needs*

🏆 Task 1.3 — Economics of Cloud Computing: Final Exam Tips

1 CapEx vs OpEx

- **CapEx (Capital Expenditure):** Upfront spending on servers, data centers.
 - **Use Case:** Traditional on-premises setup for an enterprise → high initial investment.
 - **OpEx (Operational Expenditure):** Pay-as-you-go, operational costs for running resources.
 - **Use Case:** Startups launch EC2 instances only when needed → pay hourly.
 - **Exam Tip:** *“Avoid upfront costs → OpEx / Pay-as-you-go”*
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2 AWS Pricing Fundamentals

Concept	Key Idea	Example / Exam Tip
Pay-as-you-go	Only pay for resources used	Testing workloads on EC2 or Lambda → pay only when running
Save when you reserve	Reserved Instances → commit 1–3 years → lower cost	Stable, predictable workloads → Reserved EC2 / RDS
Pay less by using more	Volume discounts for large usage	S3 storage > 100 TB → cheaper per GB

3 Total Cost of Ownership (TCO)

- **Definition:** All costs of running IT systems (hardware, software, maintenance, staffing).
 - **AWS Use Case:** Migrating ERP to AWS → reduces server, power, staff costs → lowers TCO.
 - **Exam Tip:** Questions often ask: “*Company wants to reduce overall IT costs*” → *TCO reduction via cloud.*
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4 AWS Pricing Calculator & Simple Monthly Calculator

- **Purpose:** Estimate monthly AWS costs and plan budgets.
 - **Use Case:** Startup forecasts EC2 + S3 + Lambda monthly cost using **AWS Pricing Calculator**.
 - **Exam Tip:** “*Estimate monthly cloud costs* → *think Pricing Calculator*”
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5 Cost Optimization Methods

Method	How It Saves Costs	Example / Exam Tip
Auto Scaling	Scale resources automatically → avoid paying for idle resources	E-commerce site scales EC2 during Black Friday → pay less off-peak
Right-Sizing	Match instance type to workload → avoid over-provisioning	Replace large EC2 with smaller during low usage
Spot Instances	Use spare capacity at huge discount → up to 90% savings	Batch data processing → flexible, interruptible workloads

6 Elasticity & Cost Savings

- **Definition:** Automatically scale resources up/down based on demand.
 - **Use Case:** Travel booking app scales servers during peak hours → reduces idle costs.
 - **Exam Tip:** Elasticity = **dynamic scaling** → **cost efficiency**
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⚡ Quick Memory Mnemonics

1. “O-P-R-S-T-E” → OpEx, Pay-as-you-go, Reserved, Spot, TCO, Elasticity (“OP-RSTE”)
 2. Elasticity & Auto Scaling → reduce costs dynamically
 3. Reserved = stable workloads, Spot = flexible workloads
 4. Pricing Calculator → estimate monthly cost
 5. TCO → all IT costs → AWS reduces hardware + staff + power costs
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💡 Exam Shortcut / Point of View

- Scenario mentions:
 - “Avoid upfront investment” → OpEx
 - “Scale automatically during demand spikes” → Auto Scaling / Elasticity
 - “Lower total IT costs” → TCO
 - “Predictable workloads → long-term discount” → Reserved Instances
 - “Flexible batch jobs → huge discounts” → Spot Instances

🕒 Task 1.4 — Cloud Computing Models: Final Exam Tips

1 Deployment Models

Model	Key Idea	Use Case / Example	Exam Tip
Public Cloud	Shared resources over the internet , pay-as-you-go	Startup hosts website on AWS EC2 + S3	“Internet + shared → public cloud”
Private Cloud	Dedicated to one organization , more secure	Bank storing sensitive data on AWS Outposts	“Single org → private cloud”
Hybrid Cloud	Combination of private + public cloud	Company keeps sensitive HR data private, web apps on AWS	“Mix → hybrid cloud”

Model	Key Idea	Use Case / Example	Exam Tip
Multi-Cloud (optional for exam)	Uses multiple cloud providers	Company uses AWS + Azure + GCP	<i>“Multiple vendors”</i>

2 Service Models

Model	Key Idea	AWS Example	Exam Tip
IaaS	Rent infrastructure → manage apps yourself	EC2, EBS, VPC	<i>“Rent servers/storage/networks → you manage apps”</i>
PaaS	Platform provided → deploy & run apps	AWS Elastic Beanstalk	<i>“Focus on apps, infra managed by AWS”</i>
SaaS	Fully managed software → accessible via web	Amazon WorkSpaces, AWS Chime	<i>“Use software, no setup required”</i>

3 Skills for Exam

- **Identify which deployment model fits a business scenario:**
 - Internet + shared = Public
 - Single org = Private
 - Mix of private + public = Hybrid
 - **Differentiate service models:**
 - IaaS = rent infra, manage apps
 - PaaS = platform provided, deploy apps
 - SaaS = fully managed software
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⚡ Memory Mnemonics

- **Deployment Models:** PPH (M) → **Public, Private, Hybrid (Multi-cloud)**
 - **Service Models:** I-P-S → **IaaS → PaaS → SaaS**
 - **Quick tip:** Think “rent infra → IaaS, deploy apps → PaaS, use software → SaaS”
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🧠 Exam Shortcut / Point of View

- Scenario mentions “use AWS EC2, manage apps yourself” → **IaaS**
- Scenario mentions “deploy app without managing servers” → **PaaS**
- Scenario mentions “use software like WorkSpaces or Chime” → **SaaS**
- Scenario mentions “mix of private + public cloud” → **Hybrid**
- Scenario mentions “dedicated infrastructure” → **Private**
- Scenario mentions “shared on internet” → **Public**

2 Shared Responsibility Model – Full Exam Tips

1 Definition

AWS splits **security responsibilities** into two parts:

Responsibility	Meaning
AWS (Security of the Cloud)	Protects infrastructure : data centers, hardware, network, managed services
Customer (Security in the Cloud)	Secures your data & applications : IAM, OS, app security, network configs

Mnemonic:

AWS = Infra / You = Data + Apps ☒

2 AWS Responsibility – Security of the Cloud

AWS is responsible for **everything below the hypervisor** and **managed services**:

Area	What AWS Does	Example / Use Case
Physical security	Guards, cameras, access control	Data center doors, 24/7 monitoring
Hardware	Servers, storage, networking	EC2 physical servers, S3 disks
Software	Hypervisors, virtualization	EC2 virtualization layer patched by AWS
Managed services	RDS, DynamoDB, Lambda	Patching & securing the OS for managed DBs

Exam Tip:

- Question: “Who secures the data center?” → **AWS**
 - Question: “Who patches RDS OS in managed service?” → **AWS**
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3 Customer Responsibility – Security in the Cloud

You control **what you deploy and how you configure it**:

Area	What Customer Does	Example / Use Case
Data	Classification, encryption, backups	Encrypt S3 objects, backup databases
IAM	Users, groups, roles, policies	Control who can access EC2, S3, Lambda

Area	What Customer Does	Example / Use Case
OS & Applications	Patch OS, secure applications	Update EC2 OS, secure your web apps
Network Traffic	Security Groups, NACLs, VPN	Control inbound/outbound firewall rules
Compliance	Regulatory responsibility	HIPAA, GDPR compliance for your data

Exam Tip:

- Question: “Who patches EC2 OS?” → **Customer**
- Question: “Who encrypts S3 objects?” → **Customer**
- Question: “Who configures Security Groups?” → **Customer**

4 Use Cases / Examples

1. **Web App on EC2 + S3**
 - AWS: EC2 host security, network, S3 storage
 - Customer: OS updates, IAM access, S3 encryption
2. **Serverless App (Lambda + DynamoDB)**
 - AWS: Runtime environment, DynamoDB servers, infrastructure security
 - Customer: Lambda code, API keys, IAM roles
3. **HIPAA Health App**
 - AWS: HIPAA-compliant infrastructure
 - Customer: Encrypt data, manage access, enable logging

5 Quick Recall Table – Exam Style

Question	Answer
Who secures physical servers?	AWS
Who patches EC2 OS?	Customer
Who encrypts S3 data?	Customer
Who manages IAM users & roles?	Customer
Who monitors hypervisor security?	AWS
Who configures Security Groups?	Customer

6 Final Exam Tips

- Always ask: “Is this **infrastructure** or **my data/app**?” → AWS = infra, Customer = data/app
- Most questions are **scenario-based**
- Use **mnemonics & table** to answer quickly

📌 IAM – Final Exam Tips

1️⃣ Key Concepts & Mnemonics

Mnemonic:

****U-G-R-P-M-L** → Users, Groups, Roles, Policies, MFA, Least privilege** ✅

Term	Quick Definition	Exam Tip / Use Case
Users	Individual AWS credentials	“Developer needs console login?” → User
Groups	Collection of users sharing permissions	“Multiple developers need same EC2 access?” → Group
Roles	Temporary credentials for services or users	“EC2/Lambda needs access to S3 or CloudWatch?” → Role
Policies	JSON rules to Allow/Deny actions	“Restrict S3 write access?” → Use Policy
MFA	Extra login security using app/device	“Secure root account?” → Enable MFA
Least Privilege	Give only required permissions	“Dev only needs read access?” → Apply least privilege principle

2️⃣ Real-Time Exam Use Cases

1. **Developer login to AWS console → IAM User**
 2. **Team of developers sharing permissions → IAM Group**
 3. **EC2 accessing S3 → IAM Role**
 4. **Lambda writing logs to CloudWatch → IAM Role**
 5. **Restrict root account / sensitive users → MFA**
 6. **Grant only necessary permissions → Least Privilege Principle**
 7. **Restrict temporary contractor from deleting EC2 → Policy + Least Privilege**
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3️⃣ Exam Tips

- Look for **keywords**:
 - “Temporary access” → **Role**
 - “Multiple users same permissions” → **Group**
 - “Individual login” → **User**
 - “Secure login” → **MFA**
 - “Minimal required permissions” → **Least Privilege**
 - Most IAM questions are **scenario-based**, pick the feature that **matches the scenario**.
 - Policies are **always JSON**, so when the question mentions “Allow/Deny rules” → think **Policies**.
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☑ Final Shortcut Table

Scenario / Keyword	IAM Feature
Single developer login	User
Multiple developers, same permissions	Group
EC2/Lambda temporary access	Role
Define Allow/Deny actions	Policy
Secure login	MFA
Only required permissions	Least Privilege

📌 AWS Cognito – Final Exam Tips

1️⃣ Key Points

- **User authentication & management** → Securely manage app users.
- **Sign-in / Sign-up** → Mobile & web applications.
- **Integration with social identity providers** → Google, Facebook, Amazon, SAML.

Mnemonic:

Cognito = Auth + SignIn/SignUp + Social Login ☑

2️⃣ Core Components

Component	Purpose	Exam Tip / Use Case
User Pool	User directory for authentication	App needs users to register/sign-in
Identity Pool	Provides temporary AWS credentials	Authenticated users need to access S3/DynamoDB
Federation	Social/enterprise login	Enable Google/Facebook login for app
MFA	Extra security layer	Two-factor authentication for accounts

3️⃣ Real-Time Exam Use Cases

1. Mobile app needs registration & login → **User Pool**
 2. Authenticated users need access to AWS resources → **Identity Pool**
 3. Social login (Google/Facebook) → **Federation**
 4. Extra login security → **Enable MFA**
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4️⃣ Quick Exam Table

Scenario	Cognito Feature
App user registration & login	User Pool
Temporary AWS access for users	Identity Pool
Social login integration	Federation
Two-factor authentication	MFA

5 Exam Tips

- **User Pool** → authentication
- **Identity Pool** → authorization (temporary credentials)
- **Federation** → social or enterprise login
- **MFA** → adds extra security
- Use mnemonic **UP-IP-F-M** → **User Pool, Identity Pool, Federation, MFA**

7 AWS Organizations – Final Exam Tips

1 Key Points

- **Account management & central governance** → Manage multiple AWS accounts from a single master account.
- **Consolidated billing** → Combine bills from multiple accounts for easier tracking & cost optimization.
- **Service Control Policies (SCPs)** → Set permissions guardrails for accounts within the organization.

Mnemonic:

Organizations = Accounts + Billing + Policies ☒

2 Real-Time Exam Use Cases

1. **Multiple accounts management**
 - *Scenario:* “A company has separate AWS accounts for Dev, Test, and Prod environments.”
 - *Answer:* Use **AWS Organizations** to centrally manage accounts.
2. **Centralized billing**
 - *Scenario:* “Combine billing for 5 AWS accounts to simplify cost reporting.”
 - *Answer:* Use **Consolidated Billing**.
3. **Enforcing account-level restrictions**
 - *Scenario:* “Prevent certain accounts from launching EC2 in specific regions.”
 - *Answer:* Use **Service Control Policies (SCPs)** to restrict permissions.

3 Quick Exam Table

Scenario / Keyword	AWS Organizations Feature
Manage multiple AWS accounts	Account management & central governance
Combine bills from multiple accounts	Consolidated billing
Apply guardrails / restrict permissions	Service Control Policies (SCPs)

4 Exam Tips

- Always remember **Organizations = multi-account management**.
- **SCPs do not grant permissions**, they only **limit what the account can do**.
- Consolidated billing does **not affect permissions**, only cost aggregation.
- Questions are usually **scenario-based**: “You need to control multiple accounts and restrict certain actions?” → Use **Organizations + SCPs**.

Mnemonic for exams:

Org = Accounts + Billing + SCPs ☒

2 AWS Security, Identity & Compliance Services – Final Exam Tips

1 Key Services & Purpose

Service	Purpose	Real-Time Use Case
KMS (Key Management Service)	Create & manage encryption keys, envelope encryption	Encrypt S3 objects or EBS volumes
Secrets Manager	Store DB/API keys securely	Protect database credentials used by apps
Shield (Standard & Advanced)	DDoS protection	Prevent DDoS attacks on web applications
WAF (Web Application Firewall)	Protect web apps from exploits	Block SQL injection or XSS attacks on a website
Macie	Discover sensitive data in S3	Detect PII (personal info) like credit card numbers or emails

2 Real-Time Exam Use Cases

1. Encrypt S3 objects → KMS

2. **Store API keys for applications** → **Secrets Manager**
3. **Prevent DDoS attacks on public website** → **Shield**
4. **Block malicious web traffic (SQLi/XSS)** → **WAF**
5. **Detect sensitive customer info in S3** → **Macie**

3 Quick Exam Table

Scenario / Keyword	Service
Manage encryption keys	KMS
Store secrets (DB/API keys)	Secrets Manager
Protect against DDoS	Shield
Protect web apps from exploits	WAF
Discover sensitive data	Macie

4 Exam Tips

- **KMS** = key creation & encryption
- **Secrets Manager** = store sensitive credentials securely
- **Shield** = DDoS protection
- **WAF** = application-level firewall
- **Macie** = discover sensitive data automatically in S3

Mnemonic:

K-S-S-W-M → **KMS, Secrets Manager, Shield, WAF, Macie** ✓

- Most exam questions are **scenario-based**, e.g., “Which service do you use to encrypt data at rest?” → **KMS**

7 AWS Compliance Programs – Final Exam Tips

1 Key Points

AWS Compliance Programs help **customers meet regulatory and industry standards**. AWS provides frameworks, certifications, and reports for auditing and legal compliance.

Main Programs Include:

- **HIPAA** – Healthcare data protection
- **PCI DSS** – Payment Card Industry Data Security Standard
- **SOC 1/2/3** – Audit reports for financial and operational controls
- **ISO 27001/27017/27018** – Information security standards
- **GDPR** – Data protection for EU citizens
- **FedRAMP** – US government cloud compliance

Mnemonic:

H-P-S-I-G-F → **HIPAA, PCI, SOC, ISO, GDPR, FedRAMP** ✓

2 Real-Time Exam Use Cases

1. **Healthcare app storing patient data** → Ensure **HIPAA compliance**
 2. **E-commerce app handling credit card info** → Ensure **PCI DSS compliance**
 3. **Financial organization auditing controls** → Review **SOC reports**
 4. **Global company handling EU data** → Ensure **GDPR compliance**
 5. **US federal agency using AWS** → Must meet **FedRAMP compliance**
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3 Quick Exam Table

Scenario / Keyword	Compliance Program
Healthcare data	HIPAA
Payment card info	PCI DSS
Audit of internal controls	SOC 1/2/3
Information security management	ISO 27001/27017/27018
EU citizen data protection	GDPR
US government cloud usage	FedRAMP

4 Exam Tips

- Most questions are **scenario-based**, e.g., “Which compliance standard is needed for storing patient health data?” → **HIPAA**
- Remember **mnemonic** H-P-S-I-G-F for **quick recall**
- AWS provides **pre-built compliance frameworks**, so you don’t need to configure manually for certification; just know **which applies to which scenario**

7 Data Protection & Encryption – Final Exam Tips

1 Key Points

Data protection in AWS ensures that **data at rest and in transit** is secure using **encryption and key management**.

- **Encryption at rest** → Protects stored data (S3, EBS, RDS, DynamoDB)
- **Encryption in transit** → Protects data while moving between services (HTTPS, TLS)
- **Key Management** → Use AWS **KMS** or **CloudHSM** to create and manage encryption keys
- **Secrets Management** → Use **Secrets Manager** to securely store credentials

Mnemonic:

E-K-S → **Encryption + KMS + Secrets Manager** ✓

2 Real-Time Exam Use Cases

1. **Encrypt S3 bucket data** → Use **S3 SSE (Server-Side Encryption) + KMS key**
 2. **Encrypt RDS database** → Use **KMS-managed key**
 3. **Secure API credentials** → Use **Secrets Manager**
 4. **Protect data while transferring to another region** → Use **TLS/HTTPS**
 5. **Compliance requirement for encrypted data at rest** → Enable **KMS + encryption for storage services**
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3 Quick Exam Table

Scenario / Keyword	Service / Feature
Encrypt S3 / EBS / RDS	KMS + SSE
Store DB/API secrets	Secrets Manager
Encrypt data in transit	HTTPS/TLS
Generate & manage encryption keys	KMS / CloudHSM
Protect sensitive customer data	KMS + SSE + Secrets Manager

4 Exam Tips

- **KMS** = key management for encryption
- **Secrets Manager** = securely store credentials
- **SSE** = server-side encryption for S3, EBS, RDS
- **HTTPS / TLS** = encryption in transit
- Always think “**Is it at rest or in transit?**” → choose service accordingly
- Questions often ask **which service to use for secure storage vs key management**

Mnemonic for exams:

E-K-S → **Encryption (at rest & in transit), KMS (keys), Secrets Manager (credentials)**



📌 Network Security – Final Exam Tips

1️⃣ Key Points

AWS Network Security ensures **safe and controlled network access** to your resources.

- **VPC (Virtual Private Cloud)** → Isolated network for AWS resources
- **Security Groups** → Virtual firewall for EC2 instances (stateful)
- **Network ACLs** → Subnet-level firewall (stateless)
- **VPN / Direct Connect** → Secure connection between on-premises & AWS
- **AWS WAF** → Protect web apps from exploits (SQLi, XSS)
- **AWS Shield** → DDoS protection

Mnemonic:

V-S-N-W-W → **VPC, Security Groups, Network ACLs, WAF, Shield** ☑️

2️⃣ Real-Time Exam Use Cases

1. **Isolate resources in AWS** → Use **VPC**
 2. **Control traffic to EC2 instance** → Use **Security Group**
 3. **Control traffic at subnet level** → Use **Network ACL**
 4. **Connect corporate network securely to AWS** → Use **VPN / Direct Connect**
 5. **Protect web application from SQL injection / XSS** → Use **WAF**
 6. **Prevent DDoS attacks on your website** → Use **Shield**
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3️⃣ Quick Exam Table

Scenario / Keyword	Service / Feature
Isolated network	VPC
EC2 instance traffic control	Security Group
Subnet-level firewall	Network ACL
Secure on-premises connection	VPN / Direct Connect
Protect web apps from exploits	WAF
Protect from DDoS	Shield

4️⃣ Exam Tips

- **Security Groups** = instance-level, stateful
- **Network ACLs** = subnet-level, stateless
- **VPC** = isolates network, defines IP range, subnets

- **VPN / Direct Connect** = secure hybrid network connection
- **WAF & Shield** = protect web apps from attacks (Layer 7) & DDoS
- Questions often describe **traffic rules, access control, or attack protection** → map to the correct service

Mnemonic for exams:

V-S-N-W-W → **VPC, Security Groups, Network ACLs, WAF, Shield** ☑

🔍 Security Best Practices – Final Exam Tips

1️⃣ Key Points

AWS Security Best Practices ensure **your cloud environment is secure, compliant, and resilient**.

Core Best Practices:

1. **Enable MFA** → Especially for root and privileged users
2. **Use IAM Roles instead of access keys** → Temporary credentials for services
3. **Apply least privilege principle** → Give users only the permissions they need
4. **Rotate credentials regularly** → Passwords, access keys
5. **Monitor with CloudTrail & CloudWatch** → Audit logs & metrics
6. **Encrypt sensitive data** → KMS, SSE, Secrets Manager
7. **Use network security controls** → VPC, Security Groups, NACLs, WAF, Shield

Mnemonic:

M-R-L-M-M-E-N → **MFA, Roles, Least Privilege, Monitor, Encrypt, Network security** ☑

2️⃣ Real-Time Exam Use Cases

1. **Root account protection** → Enable **MFA**
2. **EC2 instance needs S3 access** → Assign **IAM Role**
3. **Developer only needs read access** → Apply **least privilege principle**
4. **Rotate API keys every 90 days** → Follow **credential rotation**
5. **Audit user activity and API calls** → Use **CloudTrail + CloudWatch**
6. **Encrypt S3 bucket with KMS** → Protect sensitive data
7. **Restrict inbound traffic to EC2** → Use **Security Groups / NACL / WAF**

3️⃣ Quick Exam Table

Scenario / Keyword	Best Practice
Root account protection	MFA
Temporary access for services	IAM Roles
Only necessary permissions	Least Privilege
Monitor AWS usage	CloudTrail + CloudWatch
Encrypt sensitive data	KMS / SSE / Secrets Manager
Secure network traffic	Security Groups / NACL / WAF / Shield
Rotate credentials	Regular key/password rotation

4 Exam Tips

- **Always protect root account** → MFA mandatory
- **Temporary access** → Use IAM Roles, not permanent keys
- **Least privilege** principle = always give minimum required permissions
- **Logging & monitoring** → CloudTrail (audit) + CloudWatch (metrics/alarms)
- **Encrypt sensitive data** at rest and in transit
- **Network security** → combine SG, NACL, WAF, Shield for layered protection

Mnemonic for exams:

M-R-L-M-M-E-N → **MFA, Roles, Least Privilege, Monitor, Encrypt, Network security** ☑

3.1 Methods of Deploying & Operating in AWS – Cheat Sheet with Use Cases

Knowledge + Skills + Use Cases

1 Provisioning & Operating AWS Cloud

Method	How to Access	Use Case / Example
Console (GUI)	AWS Management Console	One-time setup: Launch an EC2 instance manually, check S3 bucket settings
CLI / SDK / API (Programmatic Access)	Command Line / Python Boto3 / Java SDK / REST API	Automate tasks: Start/stop EC2 instances daily, upload files to S3 programmatically

Method	How to Access	Use Case / Example
Infrastructure as Code (IaC)	CloudFormation / Terraform	Repeatable deployments: Launch multi-tier application with EC2, RDS, VPC in one script

2 Ways to Access AWS Services

Access Type	Example Use Case
Console (GUI)	Visualize resources, monitor CloudWatch dashboards, manually create IAM user
CLI	Quickly manage multiple resources via commands, e.g., deploy hundreds of EC2 instances
SDK	Integrate AWS services into your application (e.g., upload images to S3 from Python app)
API	Automate workflows, e.g., trigger Lambda via HTTP request from external system

3 Cloud Deployment Models

Model	Use Case / Example
Cloud	Entire application hosted in AWS: Website hosted on EC2 + S3 + RDS
Hybrid	On-premises database connected to AWS cloud: Use Direct Connect for data backup to S3
On-premises	Local private data center for sensitive workloads: Legacy ERP system not migrated to cloud

4 Connectivity Options

Option	Use Case / Example
AWS VPN	Securely connect office network to AWS VPC over the internet
AWS Direct Connect	High-speed private connection for large-scale data transfer to AWS
Public Internet	Access AWS services from any location without dedicated link

Exam Mnemonics / Scenario Tips

- **Access Methods:** Console, API/CLI/SDK, IaC → “**CAI**”
- **Deployment Models:** Cloud, Hybrid, On-prem → “**CHO**”
- **Connectivity Options:** VPN, Direct Connect, Internet → “**VDI**”

Scenario Example Questions:

1. “You want to launch 50 EC2 instances automatically every morning.” → **Use CLI / SDK / IaC**
2. “You need a high-speed private connection from your office to AWS.” → **Use Direct Connect**
3. “A one-time test server is needed.” → **Use Console**
4. “Backup sensitive on-premises data to AWS securely.” → **Use Hybrid + VPN / Direct Connect**

3.2 AWS Global Infrastructure – Cheat Sheet with Use Cases

Knowledge (What to memorize)

1. **AWS Regions** – Geographically isolated areas (e.g., US-East-1, Asia Pacific-South-1)
2. **Availability Zones (AZs)** – Isolated locations within a Region; independent power, networking, connectivity
3. **Edge Locations** – For caching content close to users (CloudFront)
4. **High Availability** – Ensures services remain available by using multiple AZs
5. **Multiple Regions** – For disaster recovery, business continuity, low latency, data sovereignty
6. **AWS Wavelength / Local Zones** – Extend AWS services closer to end users for ultra-low latency

Skills (What to apply in the exam)

Concept	Skill / Exam Tip	Use Case / Example
Region – AZ – Edge Location Relationship	Know the hierarchy: Region > AZ > Edge Location	EC2 launched in Singapore region (Region) across 2 AZs for HA, CloudFront caches content at nearby Edge Locations
High Availability with multiple AZs	Deploy resources in multiple AZs to avoid single points of failure	Multi-AZ RDS ensures DB availability if one AZ fails
Recognize AZ isolation	AZs are independent → failure in one AZ does not affect others	EC2 instance in AZ-1 can fail, but AZ-2 instance continues serving traffic

Concept	Skill / Exam Tip	Use Case / Example
Multiple Regions usage	Disaster recovery, business continuity, low latency, data sovereignty	Backup production database to another region for DR & compliance
Edge Locations benefits	Reduce latency, cache content globally	CloudFront serves website content quickly worldwide; Global Accelerator improves global app performance
Wavelength / Local Zones	Ultra-low latency for specific workloads	Gaming or video streaming app needing <10ms latency uses Wavelength Zones near 5G networks

Exam Mnemonics / Tips

- **Hierarchy: Region → AZ → Edge Location → “RAE”**
 - **High Availability:** Spread across **multiple AZs**
 - **Disaster Recovery / Compliance:** Use **multiple Regions**
 - **Global Content / Low Latency:** Use **Edge Locations & CloudFront**
 - **Ultra-low latency apps:** Use **Wavelength / Local Zones**
-

Scenario Example Questions

1. “Ensure a web application remains available if one data center fails.” → **Deploy across multiple AZs**
2. “Deliver static website content globally with low latency.” → **Use CloudFront Edge Locations**
3. “Meet data residency requirements in another country.” → **Use another AWS Region**
4. “Ultra-low latency for a 5G mobile app.” → **Use Wavelength / Local Zones**

3.3 AWS Compute Services – Complete Table

Service / Concept	Purpose / Knowledge	Exam Tip / Skill	Use Case / Example
EC2 (Elastic Compute Cloud)	Virtual servers in AWS	Choose instance type based on workload	Launch web servers, backend apps, test servers
EC2 Instance Types	Compute-optimized, Storage-optimized, Memory-optimized, General-purpose	Match instance type to workload	CPU-heavy → compute-optimized; DB → memory-optimized; Data warehouse → storage-optimized
Containers: ECS	AWS-managed container orchestration	Simple containerized apps	Run microservices or container apps without Kubernetes
Containers: EKS	Kubernetes-managed container orchestration	Use for complex apps with K8s	Run multi-container apps with Kubernetes orchestration
Serverless: Lambda	Run code without servers	Event-driven workloads	Process images when uploaded to S3, run backend APIs
Serverless: Fargate	Run containers serverless	Avoid managing EC2 for containers	Run containerized apps automatically without provisioning servers
Auto Scaling	Automatically adjusts number of instances	Provides elasticity	Scale web servers up during traffic spikes, scale down at night
Load Balancers (ELB)	Distribute traffic across multiple instances/AZs	Ensure high availability and fault tolerance	Direct incoming web requests to multiple EC2 instances across AZs
Elastic Beanstalk	Platform as a Service (PaaS) deployment	Simplify app deployment	Deploy a web app with pre-configured EC2, RDS, and scaling
Lightsail	Simplified VPS hosting	Quick small apps or learning	Launch a personal blog or test environment easily

Quick Mnemonics / Exam Tips

- **Compute Options:** “EC2 → Containers → Serverless → PaaS/VPS” → ECSPL
- **Scaling & Availability:** Auto Scaling + Load Balancer → “Always Available”
- **Scenario Trick:** Ask yourself:
 - *Do I need a server?* → EC2 / Lightsail

- *Containers?* → ECS / EKS / Fargate
- *No server / event-driven?* → Lambda / Fargate
- *Scale automatically?* → Auto Scaling + ELB

💡 **Scenario Examples for Exam:**

1. **Batch CPU-heavy job** → Compute-optimized EC2
2. **Containerized app without managing servers** → Fargate
3. **Web app with traffic spikes** → EC2 + Auto Scaling + ELB
4. **Small test website quickly** → Lightsail
5. **Image processing on S3 upload** → Lambda

3.4 AWS Database Services – Complete Table

Service / Concept	Purpose / Knowledge	Exam Tip / Skill	Use Case / Example
Amazon RDS (Relational Database Service)	Managed relational database	Use when you need a managed SQL DB	MySQL, PostgreSQL, Oracle, SQL Server databases for web apps
Amazon Aurora	High-performance, MySQL & PostgreSQL-compatible DB	Choose for high availability & performance	E-commerce database needing auto-scaling & fault tolerance
Amazon DynamoDB	Managed NoSQL database	Use for key-value or document DB, serverless scaling	Store user profiles, session data, gaming leaderboards
Amazon ElastiCache	In-memory database / caching	Use for ultra-fast data retrieval	Cache frequently accessed DB queries, session store, leaderboard data
Amazon Redshift	Data warehousing	Use for analytics & reporting	Analyze large datasets, BI dashboards
AWS Database Migration Service (DMS)	Migrate databases to AWS	Use to move on-prem or cloud DB to AWS	Migrate Oracle DB to Amazon RDS with minimal downtime
AWS Schema Conversion Tool (SCT)	Convert database schema	Use when migrating between DB engines	Convert Oracle schema to PostgreSQL for migration

Key Skills / Exam Tips

- **Choose DB type based on workload:**
 - Relational → RDS / Aurora
 - NoSQL → DynamoDB
 - In-memory → ElastiCache
 - Data warehouse → Redshift
 - **Managed vs EC2-hosted:** Prefer **AWS-managed services** unless you need full control
 - **Migration tools:** DMS + SCT for moving DBs safely
 - **High Availability:** Use Multi-AZ RDS or Aurora clusters for fault tolerance
-

Scenario Examples

1. *“Need a fully managed MySQL database for a web app.”* → **RDS**
 2. *“Ultra-fast caching for a high-traffic app.”* → **ElastiCache**
 3. *“Key-value storage for user sessions.”* → **DynamoDB**
 4. *“Data warehouse for analytics on large datasets.”* → **Redshift**
 5. *“Migrate on-prem Oracle DB to AWS with minimal downtime.”* → **DMS + SCT**
 6. *“High-performance relational DB with auto-scaling & fault tolerance.”* → **Aurora**
-

💡 **Mnemonic for Exam:**

“RRDDEMS” → **RDS, Redshift, DynamoDB, ElastiCache, Migration Service (DMS), SCT**

3.5 AWS Network Services – Complete Table

Service / Concept	Purpose / Knowledge	Exam Tip / Skill	Use Case / Example
VPC (Virtual Private Cloud)	Isolated virtual network in AWS	Know components: subnets, route tables, gateways	Create secure private network for EC2, RDS, Lambda
Subnets	Segment VPC into smaller networks	Public vs private subnets	Public subnet → web servers; Private subnet → DB servers
Internet Gateway (IGW)	Enable VPC resources to access the internet	Connect VPC to internet	Public EC2 instances access web & receive traffic
NAT Gateway / NAT Instance	Allow private subnet to access internet	Secure outbound internet access	DB servers in private subnet download updates
Security Groups	Virtual firewall for instances	Control inbound/outbound traffic	Allow HTTP/HTTPS to web servers, block unwanted traffic
Network ACLs (NACLs)	Firewall at subnet level	Stateless rules for extra security	Block traffic from specific IP ranges at subnet level
Route 53	DNS & routing service	Direct traffic & manage domains	Map domain names to EC2, S3, or external endpoints
Edge Services (CloudFront, Global Accelerator)	Content delivery & latency optimization	Deliver content faster, low-latency apps	CloudFront → static website caching; Global Accelerator → improve app performance globally
AWS VPN	Secure connection from on-premises to AWS	Encrypted internet-based connection	Connect office network to AWS private VPC
AWS Direct Connect	Dedicated private connection to AWS	High-speed, low-latency, secure	Large-scale data transfer or hybrid cloud setups

Key Skills / Exam Tips

- **VPC Components:** Subnets, IGW, NAT, Route Tables → secure & route traffic correctly
- **Security Layers:** Security Groups → instance-level firewall; NACLs → subnet-level firewall
- **Traffic Optimization:** Edge services → CloudFront & Global Accelerator
- **Connectivity Options:** VPN → encrypted over internet; Direct Connect → private dedicated line

Scenario Examples

1. “Web servers need internet access; databases should remain private.” → **Public subnet for EC2, Private subnet for RDS**
 2. “Securely connect office network to AWS.” → **AWS VPN**
 3. “Transfer large data regularly with low latency and high speed.” → **Direct Connect**
 4. “Deliver static website content globally with low latency.” → **CloudFront**
 5. “Map my domain to EC2 web server.” → **Route 53**
 6. “Restrict traffic to certain IP addresses at subnet level.” → **NACLs**
 7. “Control traffic for a specific EC2 instance.” → **Security Groups**
-

💡 **Mnemonic for Exam:**
“VSPR-EN-DD” → VPC, Subnet, Route tables, IGW, Edge, NACL/Security, Direct Connect/DNS”

3.6 AWS Storage Services – Complete Table

Service / Concept	Purpose / Knowledge	Exam Tip / Skill	Use Case / Example
Amazon S3 (Simple Storage Service)	Object storage in the cloud	Store unstructured data, use different storage classes	Store images, videos, backups; S3 Standard for frequently accessed, S3 Glacier for archival
S3 Storage Classes	Standard, Intelligent-Tiering, Glacier, Glacier Deep Archive	Choose based on access frequency and cost	S3 Standard → web app content, Glacier → long-term backup
Amazon EBS (Elastic Block Store)	Block storage for EC2	Attach volumes to EC2 instances, persistent storage	Store OS and application data, high IOPS workloads
Instance Store	Temporary storage for EC2	Ephemeral, deleted when instance stops	Cache, temporary files, scratch data
Amazon EFS (Elastic File System)	Managed file storage	Shared file system across multiple EC2 instances	Web server clusters, shared configuration files
Amazon FSx	Managed file storage for Windows / Lustre	Choose based on file system requirements	FSx for Windows → SMB file shares; FSx for Lustre → high-performance computing
AWS Storage Gateway	Hybrid cloud storage	Cached or tape-based file system integration	Backup on-premises files to S3, low-latency cache

Service / Concept	Purpose / Knowledge	Exam Tip / Skill	Use Case / Example
Lifecycle Policies	Automate moving objects between S3 classes	Cost optimization	Move S3 objects to Glacier after 30 days
AWS Backup	Centralized backup service	Backup EC2, RDS, DynamoDB, EFS	Schedule automated backups, compliance and disaster recovery

Key Skills / Exam Tips

- **Object vs Block vs File Storage:**
 - Object → S3, Glacier (web, archival)
 - Block → EBS, instance store (EC2 disks)
 - File → EFS, FSx (shared file systems)
 - **Lifecycle Policies:** Automate cost optimization for S3
 - **AWS Backup:** Centralized, supports multiple AWS services
 - **Temporary vs Persistent Storage:** Instance store is ephemeral, EBS/EFS is persistent
-

Scenario Examples

1. “Store user-uploaded images for a web app.” → **S3 Standard**
 2. “Archive old data cost-effectively.” → **S3 Glacier / Glacier Deep Archive**
 3. “High IOPS disk for EC2 database.” → **EBS**
 4. “Temporary cache for EC2 app.” → **Instance Store**
 5. “Shared configuration files for multiple EC2 instances.” → **EFS**
 6. “Windows file shares for an app cluster.” → **FSx for Windows**
 7. “Automate moving older files to cheaper storage.” → **S3 Lifecycle Policies**
 8. “Centralized backup of multiple AWS services.” → **AWS Backup**
-

💡 **Mnemonic for Exam:**

“SOF-LB” → S3, Object classes, EBS, File storage (EFS/FSx), Lifecycle, Backup

Scenario	AWS Service
Train ML model for demand forecasting	SageMaker
Customer service chatbot	Lex
Enterprise document search	Kendra
Detect objects or faces in images	Rekognition
Convert text to speech	Polly
Analyze sentiment from reviews	Comprehend
Translate website content	Translate
Query S3 logs using SQL	Athena
Real-time clickstream analytics	Kinesis
ETL data into warehouse	Glue
Create dashboards for management	QuickSight

Mnemonic for Exam

AI/ML Services: “Sage Lex Kendra Rek Polly Com Trans” → SLKR-PCT
Analytics Services: “Athena Kinesis Glue QuickSight” → AKGQ

💡 Pro Tip:

- **Keywords in scenario questions** often point to the right service:
 - “Real-time data” → Kinesis
 - “Serverless SQL query” → Athena
 - “Dashboards & BI” → QuickSight
 - “Train or deploy ML models” → SageMaker
 - “Chatbot” → Lex
 - “Image recognition” → Rekognition

3.8 Other AWS Services – Complete Table

Service Category	Service	Purpose / Knowledge	Exam Tip / Skill	Use Case / Example
Application Integration	EventBridge	Event-driven architecture, route events	Use for serverless event orchestration	Trigger Lambda or EC2 on events from apps
	SNS (Simple Notification Service)	Pub/Sub messaging, alerts	Push notifications to multiple subscribers	Send SMS/email alerts for system events

Service Category	Service	Purpose / Knowledge	Exam Tip / Skill	Use Case / Example
Business Applications	SQS (Simple Queue Service)	Message queuing	Decouple microservices & applications	Queue tasks for background processing
	Amazon Connect	Cloud-based contact center	Customer support & call management	Call center for customer service
	Amazon SES (Simple Email Service)	Send transactional emails	Bulk email sending or notifications	Marketing emails, password reset emails
Customer Engagement	AWS Activate	Startup support & resources	Use for startup credits & guidance	Provide startups AWS credits & tools
	AWS IQ	Connects customers with certified experts	Hire experts for projects	Outsource AWS consulting tasks
	AWS Managed Services (AMS)	Managed AWS environment	Simplify operational management	Managed AWS accounts for enterprises
Developer Tools	AWS Support	AWS technical support	Access support plans	Choose Basic, Developer, Business, Enterprise support
	AppConfig	App configuration management	Deploy app configurations safely	Update feature flags without redeploying apps
	Cloud9	Cloud-based IDE	Code in browser	Develop Lambda or EC2 apps without local setup
	CloudShell	Command-line environment	Browser-based shell for AWS CLI	Quickly run CLI commands without setup
	CodeArtifact	Package management	Store & share code packages	Host private npm/maven repositories
	CodeBuild	Build automation	Compile and test code	Build Docker images for deployment
	CodeCommit	Source control	Git-based repository in AWS	Store application code securely
	CodeDeploy	Automated deployment	Deploy apps to EC2, Lambda, or on-prem	CI/CD deployments
	CodePipeline	CI/CD workflow	Automate build-test-deploy	Create pipeline for DevOps workflows
	CodeStar	DevOps project management	Unified interface for DevOps	Manage DevOps projects from one dashboard

Service Category	Service	Purpose / Knowledge	Exam Tip / Skill	Use Case / Example
End-User Computing	X-Ray	Debug & trace apps	Analyze distributed apps	Find bottlenecks in microservices
	AppStream 2.0	Stream applications to users	Access apps without local installation	Stream desktop apps to remote users
	WorkSpaces	Virtual desktops	Provide cloud desktops to users	Remote work desktops for employees
Frontend Web & Mobile	WorkSpaces Web	Browser-based desktops	Access desktops in browser	Access corporate desktops without client
	Amplify	Build & deploy web/mobile apps	Fullstack development & hosting	Host React or Angular web apps
	AppSync	GraphQL API service	Manage data for frontend apps	Real-time mobile app data synchronization
IoT Services	AWS IoT Core	Connect & manage IoT devices	Device communication & data collection	Smart home sensors, industrial IoT
	AWS IoT Greengrass	Local compute for IoT	Run IoT apps offline	Edge processing for connected devices

Key Skills / Exam Tips

- **Messaging & Alerts:** EventBridge (event routing), SNS (notifications), SQS (queueing)
- **Business Apps:** Connect (call center), SES (emails)
- **Customer Support:** AWS Support, Activate, IQ, Managed Services
- **Developer Tools:** CodeSuite (CodeBuild, CodeCommit, CodeDeploy, CodePipeline, CodeStar), AppConfig, X-Ray
- **End-User Computing:** AppStream, WorkSpaces, WorkSpaces Web
- **Frontend / Mobile:** Amplify, AppSync
- **IoT Management:** IoT Core, IoT Greengrass

Scenario-Based Examples

Scenario	AWS Service
Send SMS or email alerts	SNS
Queue tasks for background processing	SQS
Event-driven app workflow	EventBridge
Cloud-based call center	Connect

Scenario	AWS Service
Send bulk transactional emails	SES
Develop cloud apps in browser	Cloud9 / CloudShell
CI/CD automation	CodePipeline + CodeBuild + CodeDeploy
Trace performance of microservices	X-Ray
Stream desktop apps to users	AppStream 2.0
Provide virtual desktops	WorkSpaces
Build & host web/mobile apps	Amplify
Manage real-time mobile app data	AppSync
Connect IoT devices to cloud	IoT Core
Process IoT data at edge	IoT Greengrass

💡 **Mnemonic for Exam: “M-C-D-D-E-F-I”**

- **M** → Messaging (SNS, SQS, EventBridge)
- **C** → Customer / Business (Connect, SES, Support, Activate, IQ)
- **D** → Developer Tools (AppConfig, Cloud9, CloudShell, CodeSuite, X-Ray)
- **E** → End-User Computing (AppStream, WorkSpaces, WorkSpaces Web)
- **F** → Frontend / Mobile (Amplify, AppSync)
- **I** → IoT (IoT Core, IoT Greengrass)

1 AWS Pricing Models with Use Cases

Pricing Model	Description	Use Case / Example	Exam Tip
On-Demand Instances	Pay-as-you-go; no long-term commitment.	A startup launches a new app for 2 weeks to test demand. They don't know the traffic yet.	Flexible, no upfront cost. "Short-term / unpredictable workload" → On-Demand
Reserved Instances (RI)	Commit 1 or 3 years, get discount.	A company has a production database running 24/7 for a year. They want predictable costs.	Steady long-term workloads. "Commitment, predictable" → RI
Savings Plans	Flexible discount if you commit to consistent usage (compute hours).	A company runs multiple EC2 instances in different regions constantly. Wants automatic discount.	Similar to RI but flexible. "Flexible long-term discount" → Savings Plan
Spot Instances	Buy unused capacity at huge discount, can be interrupted.	A media company renders 3D animation frames overnight. Job can stop and resume.	Cheap, interruptible workloads. "Batch processing / fault-tolerant" → Spot
Dedicated Hosts	Physical EC2 server fully dedicated to you.	A bank needs to run a legacy licensed database that must run on a dedicated physical server.	Compliance / license requirement. "Physical server" → Dedicated Host
Capacity Reservations	Reserve capacity in a specific AZ to guarantee availability.	A gaming company launches a new multiplayer game. Must guarantee server availability during launch.	Guarantee capacity in high-demand region. "Critical workload, guaranteed AZ" → Capacity Reservation

💡 Memory Hack for Exam

Think "O-R-S-S-D-C" for **On-Demand, Reserved, Savings, Spot, Dedicated, Capacity**.

- **O** → Short-term, flexible → On-Demand
- **R** → Long-term steady → Reserved
- **S** → Flexible long-term discount → Savings
- **S** → Cheap, interruptible → Spot
- **D** → Physical / compliance → Dedicated Host
- **C** → Guaranteed AZ capacity → Capacity Reservation

2 Total Cost of Ownership (TCO) & Cost Estimation

AWS provides tools and concepts to **compare cloud costs with on-premises** and plan your budgets effectively.

A. Total Cost of Ownership (TCO)

Definition:

- TCO is a **framework to calculate the total cost of running IT infrastructure** over time, including hardware, software, labor, and other operational expenses.
- Helps organizations **justify moving to AWS** by showing potential savings.

Components of TCO:

1. **Hardware costs** – servers, storage, network equipment
2. **Software costs** – OS licenses, database licenses
3. **Facilities costs** – power, cooling, space
4. **Labor costs** – system administrators, maintenance
5. **Operational costs** – monitoring, backups, updates

Exam Tip:

- Question clue: “*Estimate cost savings vs on-premises*” → Think **TCO Calculator**.

Use Case Example:

- A company runs 10 on-premises servers and wants to migrate to AWS.
 - They use the **AWS TCO Calculator** to compare:
 - On-premises cost: \$120,000/year
 - AWS cost: \$70,000/year
 - Result: **\$50,000 annual savings**, showing value of moving to cloud.
-

B. AWS Pricing Calculator

Definition:

- Online tool to **estimate monthly cost of AWS resources before deployment**.
- Lets you configure EC2, S3, RDS, Lambda, and more.

Features:

- Supports **multiple services in one estimate**
- Allows **region selection** (pricing differs by region)
- Shows **monthly and yearly cost breakdown**

Exam Tip:

- Question clue: “*Estimate future cost before deployment*” → **AWS Pricing Calculator**.

Use Case Example:

- A startup plans to deploy a web app with:
 - 3 EC2 instances (t3.medium)
 - 200 GB S3 storage
 - 20 GB RDS database
- Using the **AWS Pricing Calculator**, they see the cost = **\$120/month**, helping plan the budget.

C. Cost Estimation Tips

1. **Region matters:** Costs differ between regions (e.g., US East vs Asia Pacific).
2. **Service type matters:** EC2 instance type, storage class, data transfer all affect cost.
3. **Estimate before deployment:** Prevent surprises on billing.
4. **Combine with Savings Plans or Reserved Instances:** Reduce projected costs.

Exam Memory Trick:

- “**TCO = Compare on-premises; Pricing Calculator = Plan cloud cost**”

☒ Quick Cheat for Exam:

Tool	Purpose	Use Case
TCO Calculator	Compare cloud vs on-premises	Show management potential cost savings
AWS Pricing Calculator	Estimate future AWS cost	Budget planning before launching services

3 Billing & Cost Management Tools

AWS provides multiple tools to **track, manage, and optimize costs**. These are frequently tested in the exam.

A. AWS Cost Explorer

Definition:

- Tool to **visualize and analyze your AWS costs and usage over time**.
- Shows trends, identifies **cost spikes**, and helps with **forecasting**.

Use Case:

- A company notices a sudden increase in EC2 costs last month.
- Using **Cost Explorer**, they find one instance type was over-provisioned.
- They resize the instance to reduce costs.

Exam Tip:

- Question clue: “*Analyze trends or visualize past costs*” → **Cost Explorer**
-

B. AWS Budgets

Definition:

- Tool to **set custom budgets** and receive **alerts when usage or costs exceed thresholds**.

Use Case:

- A team sets a **monthly EC2 budget of \$500**.
- When usage reaches \$450, AWS sends an email alert.
- Prevents surprise bills.

Exam Tip:

- Question clue: “*Alert me when cost exceeds limit*” → **AWS Budgets**
-

C. AWS Billing Dashboard

Definition:

- Central dashboard to **view bills, payments, and account details.**

Use Case:

- A company wants to **download last month's invoice.**
- They navigate to **Billing Dashboard** → Payments → Download Invoice.

Exam Tip:

- Question clue: *"Where do you manage payments or invoices?"* → Billing Dashboard
-

D. AWS Cost and Usage Report (CUR)

Definition:

- **Detailed CSV report** of AWS costs and usage.
- Can be imported into **Athena, Redshift, or QuickSight** for detailed analysis.

Use Case:

- Finance team needs **monthly departmental cost allocation.**
- CUR provides line-item detail for each resource and account.

Exam Tip:

- Question clue: *"Detailed line-item report for billing"* → CUR
-

E. AWS Cost Anomaly Detection

Definition:

- Uses **machine learning** to detect **abnormal spending patterns.**

Use Case:

- A sudden spike in S3 storage costs triggers an alert from **Cost Anomaly Detection.**
- Team identifies an unattended backup script running multiple times.

Exam Tip:

- Question clue: *"Detect unusual AWS spending automatically"* → Cost Anomaly Detection
-

F. AWS Cost Categories

Definition:

- Organize costs by **projects, business units, or teams**.
- Makes reporting **easier for chargeback or showback**.

Use Case:

- A company tags EC2 and S3 resources by **Project A** and **Project B**.
- Cost Categories show **Project A** spent **\$2,000** and **Project B** **\$3,500**.

Exam Tip:

- Question clue: “*Organize costs by project or team*” → Cost Categories

☒ Quick Exam Cheat Table

Tool	Purpose	Exam Use Case
Cost Explorer	Visualize & analyze trends	Find cost spikes & trends
Budgets	Alerts on threshold	Prevent overspending
Billing Dashboard	Manage invoices & payments	Download invoices & view bills
Cost & Usage Report (CUR)	Detailed line-item report	Departmental cost allocation
Cost Anomaly Detection	Detect unusual spending	Identify unexpected cost spikes
Cost Categories	Organize costs by project/team	Chargeback/showback reporting

AWS Support Plans – 5 Types

Support Plan	Description	Use Case / Example	Exam Tip
Basic	Free plan; includes 24/7 access to customer service, AWS documentation, whitepapers, and AWS Trusted Advisor checks .	A student exploring AWS or a startup testing services.	Free plan, documentation only.

Support Plan	Description	Use Case / Example	Exam Tip
Developer	Includes all Basic features + business hours email support.	A development team testing a new app, non-critical workloads.	“Business hours email support” → Developer
Business	Includes Developer features + 24/7 phone, chat, and email support, Trusted Advisor full checks, and access to support API.	Mid-sized company running production workloads needing 24/7 guidance.	“Production workload, 24/7 phone/chat/email” → Business
Enterprise On-Ramp	All Business features + limited access to a Technical Account Manager (TAM) and proactive guidance , for companies scaling into enterprise workloads.	Company moving from Business to Enterprise scale, not yet full enterprise.	“Stepping stone to Enterprise, partial TAM” → Enterprise On-Ramp
Enterprise	All Enterprise On-Ramp features + dedicated TAM, Concierge Support, Infrastructure Event Management, proactive guidance, and rapid response for critical workloads.	Large enterprise with mission-critical workloads needing proactive 24/7 support.	“Dedicated TAM + proactive guidance for critical workloads” → Enterprise

💡 Memory Tip

Think “**B-D-B-EO-E**” = **B**asic, **D**eveloper, **B**usiness, **E**nterprise **O**n-Ramp, **E**nterprise

- **Basic:** Free, docs
 - **Developer:** Business hours email
 - **Business:** 24/7 support for production
 - **Enterprise On-Ramp:** Partial TAM, for scaling
 - **Enterprise:** Full TAM + proactive + mission-critical
-

Quick Use Case Summary

- **Basic:** Student / startup testing services
- **Developer:** Development/test workloads
- **Business:** Production workloads needing 24/7 support
- **Enterprise On-Ramp:** Scaling enterprise, partial TAM
- **Enterprise:** Mission-critical, large-scale enterprise workloads

5 Cost Optimization Strategies

AWS provides multiple strategies to **reduce costs** and **maximize efficiency** while running workloads in the cloud.

A. Right-Sizing Resources

Definition:

- Choose the **appropriate instance type, size, and storage** based on workload needs.

Use Case:

- A company has an EC2 instance running at only **20% CPU usage** constantly.
- They **downgrade** to a smaller instance → cost decreases.

Exam Tip:

- Question clue: *“Reduce cost by matching resources to actual usage”* → Right-Sizing
-

B. Use of Reserved Instances / Savings Plans

Definition:

- Commit to **long-term usage** to get **discounts over On-Demand pricing**.

Use Case:

- A production database runs 24/7 → use **Reserved Instance** → save up to 75%.
- Flexible workloads → use **Savings Plans** for automatic discount.

Exam Tip:

- Question clue: *“Long-term predictable usage, get cost savings”* → RI or Savings Plan
-

C. Use of Spot Instances

Definition:

- Buy **unused EC2 capacity at up to 90% discount**, suitable for fault-tolerant workloads.

Use Case:

- A media company renders videos overnight → uses **Spot Instances** → huge savings.

Exam Tip:

- Question clue: “*Cheap but interruptible workloads*” → Spot
-

D. Turn Off Unused Resources

Definition:

- Stop EC2 instances, detach unused EBS volumes, delete idle resources.

Use Case:

- Development team forgets running a test EC2 instance → stops it → cost saved.

Exam Tip:

- Question clue: “*Reduce costs by eliminating idle resources*” → Stop/terminate unused resources
-

E. Use Trusted Advisor

Definition:

- AWS **advisory service** that provides recommendations for:
 - Cost optimization
 - Performance improvement
 - Security checks
 - Fault tolerance

Use Case:

- Trusted Advisor shows **underutilized EC2 instances** → company downsizes → reduces cost.

Exam Tip:

- Question clue: “*AWS tool recommends cost-saving actions*” → Trusted Advisor
-

F. Choose Appropriate Storage Class

Definition:

- AWS S3 offers multiple **storage classes** based on access frequency:
 - **S3 Standard:** Frequent access
 - **S3 Intelligent-Tiering:** Automatic optimization
 - **S3 Glacier / Deep Archive:** Long-term archival → cheaper

Use Case:

- Logs are rarely accessed → move to **Glacier** → reduces storage cost.

Exam Tip:

- Question clue: “*Store infrequently accessed data cheaply*” → Glacier / Deep Archive
-

G. Monitor & Automate

- Use **AWS CloudWatch & Budgets** to monitor costs.
- Use **auto-scaling** to dynamically match resources with demand.

Use Case:

- Auto Scaling increases EC2 instances only during peak traffic → reduces idle costs.

Exam Tip:

- Look for clues like: “*Automatically adjust resources to meet demand*” → Auto Scaling
-

☒ Quick Exam Cheat Table – Cost Optimization

Strategy	Purpose	Exam Use Case
Right-Sizing	Match resources to workload	EC2 underutilized → smaller instance
RI / Savings Plan	Discount for long-term usage	Production DB runs 24/7
Spot Instances	Cheap, interruptible workloads	Video rendering / batch jobs
Turn Off Unused	Stop idle resources	Dev/test EC2 stopped
Trusted Advisor	Recommendations	Shows underutilized resources
Storage Classes	Optimize S3 cost	Logs → Glacier / Deep Archive
Monitor & Automate	Auto scale to demand	Auto Scaling during peak traffic

6 Billing Concepts

AWS billing concepts help you **understand how you pay for AWS services**, track usage, and manage costs.

A. Pay-as-You-Go

Definition:

- You pay **only for the resources you use**, with no upfront commitment.

Use Case:

- A startup launches a new app for 2 weeks → pays only for the EC2, S3, and Lambda resources consumed.

Exam Tip:

- Question clue: “*No upfront cost, flexible payment*” → Pay-as-You-Go
-

B. CapEx vs OpEx

Definition:

- **CapEx (Capital Expenditure):** Large upfront costs for physical hardware.
- **OpEx (Operational Expenditure):** Pay for resources as they are used, like AWS services.

Use Case:

- Company moves from on-premises data center (**CapEx**) to AWS (**OpEx**) → no upfront hardware cost, pay monthly.

Exam Tip:

- Question clue: “*Cloud reduces upfront cost*” → OpEx
-

C. Free Tier

Definition:

- AWS provides **limited free usage** for 12 months or always free services.

Use Case:

- A student launches an EC2 t2.micro instance and stores 5GB in S3 → free within Free Tier limits.

Exam Tip:

- Question clue: *“Free limited usage for learning or testing”* → Free Tier
-

D. Tagging for Cost Allocation

Definition:

- Assign **tags (Key-Value pairs)** to resources to **track costs by project, department, or team.**

Use Case:

- Company tags EC2 instances by **Project A** and **Project B.**
- Monthly cost report shows **Project A = \$2,000, Project B = \$3,500.**

Exam Tip:

- Question clue: *“Organize and allocate costs by project or team”* → Tagging
-

E. Consolidated Billing (AWS Organizations)

Definition:

- Combine multiple AWS accounts under **one paying account.**
- Offers **volume discounts** and **simplified billing.**

Use Case:

- Company has 5 AWS accounts → combines them under **one payer account** → total cost is billed centrally.

Exam Tip:

- Question clue: *“Combine multiple accounts for a single bill”* → Consolidated Billing
-

Concept	Definition	Use Case	Exam Tip
Pay-as-You-Go	Pay only for what you use	Startup using EC2 2 weeks	Flexible, no upfront cost
CapEx vs OpEx	Upfront vs operational cost	On-premises → AWS	Cloud shifts to OpEx
Free Tier	Limited free usage	Student uses t2.micro + 5GB S3	Free for learning/testing
Tagging	Track costs by project/team	Project A = \$2000	Allocate costs by tag
Consolidated Billing	Centralized billing	5 accounts → 1 payer	Combine accounts, volume discounts

7 AWS Marketplace Billing

AWS Marketplace is an **online store** that offers **third-party software and services** that run on AWS. Billing is **integrated with your AWS account**, making it easier to manage costs.

A. Third-Party Software Billing

Definition:

- Pay for **software subscriptions** or **services from third-party vendors** through AWS.
- Charges appear on your **AWS monthly bill**.

Use Case:

- A company subscribes to **Splunk** for log analysis via AWS Marketplace.
- Monthly subscription cost is **automatically billed** through AWS.

Exam Tip:

- Question clue: *“Pay third-party software via AWS account”* → Third-Party Software Billing
-

B. Bring Your Own License (BYOL)

Definition:

- Use your **existing software licenses** with AWS services.
- Helps **reduce cost** if you already own licenses.

Use Case:

- A company has existing **Windows Server licenses** → uses BYOL to run EC2 Windows instances → no extra license cost.

Exam Tip:

- Question clue: “*Use existing licenses in AWS*” → BYOL

C. Software Subscriptions

Definition:

- Some software in AWS Marketplace is **billed hourly, monthly, or annually**.
- Integrated with AWS billing and **easy to track**.

Use Case:

- Company subscribes to a **security tool** on AWS Marketplace → billed monthly.

Exam Tip:

- Question clue: “*Software billing integrated with AWS*” → Software Subscription

☒ Quick Exam Cheat Table – AWS Marketplace Billing

Feature	Definition	Use Case	Exam Tip
Third-Party Software	Software billed via AWS	Splunk subscription	Pay via AWS account
Bring Your Own License (BYOL)	Use existing licenses	Windows Server license	Use your own license
Software Subscription	Hourly/monthly/annual billing	Security tool	Software billed through AWS

💡 Domain 4 Summary – Cheat Sheet

- **Pricing Models:** On-Demand, RI, Savings Plan, Spot, Dedicated, Capacity
- **TCO & Cost Estimation:** TCO Calculator, Pricing Calculator
- **Billing & Cost Management Tools:** Cost Explorer, Budgets, Billing Dashboard, CUR, Cost Anomaly Detection, Cost Categories

- **Support Plans:** Basic, Developer, Business, Enterprise On-Ramp, Enterprise
- **Cost Optimization:** Right-Sizing, Spot/RI/Savings Plan, Turn off unused, Trusted Advisor, Storage Classes, Auto Scaling
- **Billing Concepts:** Pay-as-you-go, CapEx vs OpEx, Free Tier, Tagging, Consolidated Billing
- **AWS Marketplace Billing:** Third-party software, BYOL, Software subscriptions